

PANKAJ KUMAR JANGID

Current position: PhD Research Scholar, IISER Pune India
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SUMMARY

Ph.D. researcher in theoretical chemistry specializing in stochastic modeling, statistical mechanics, and computational chemistry. My research develops physically interpretable models of catalytic reactions, molecular transport, and nonequilibrium systems. I am interested in integrating mechanistic modeling with modern machine learning to build physically grounded methods for predictive chemistry and chemical discovery.

RESEARCH INTEREST

AI for Chemistry, Statistical Mechanics, Catalysis, Chemical Kinetics, Reaction Mechanisms, Physics-Informed Machine Learning, Stochastic Thermodynamics

RESEARCH EXPERIENCE

- 2021–Present **Ph.D. Researcher**, Indian Institute of Science Education and Research (IISER) Pune
- Developed theoretical and stochastic models to investigate catalytic reaction networks, nonequilibrium transport, and molecular dynamics.
 - Applied statistical mechanics and reaction kinetics to uncover microscopic mechanisms governing complex chemical systems.
 - Designed high-performance computational frameworks in C++ for stochastic simulations and quantitative validation of theoretical predictions.
- 2020–2021 **Master's Thesis Researcher**, Indian Institute of Science Education and Research (IISER), Mohali
- Developed numerical methods to study quantum resonance states using the Siegert Pseudostate method, and analyzed resonance tunneling through phase-space distributions.

EDUCATION

- 2021 - present PhD (Chemical Physics) at **Indian Institute of Science Education and Research (IISER) Pune**
Thesis: Stochastic Dynamics and Nonequilibrium Phenomena in Complex Biological and Catalytic Systems
- 2016 - 2021 BS-MS Degree in Chemistry at **Indian Institute of Science Education and Research (IISER) Mohali**
Thesis: Revisiting Siegert Pseudostates Method for Calculation of Resonances

SELECTED MACHINE LEARNING PROJECTS

Physics-Informed Neural Networks for Non-Markovian Diffusion

 [GitHub](#)

- Developed a physics-informed neural network (PyTorch) to solve memory-dependent diffusion equations with temporal convolution kernels.

- Learned both the concentration field and unknown memory kernel directly from sparse and noisy observations.
- Implemented the complete framework from scratch, including automatic differentiation, quadrature-based memory integrals, custom loss functions, and training pipeline.

Anomalous Hurst Coefficient Prediction using Deep Sequence Models

 [GitHub](#)

- Developed deep learning models for estimating the Hurst exponent of fractional Brownian motion trajectories.
- Implemented simulation pipelines (Davies–Harte, Hosking, Cholesky), feature-based baselines, Bidirectional LSTM, and Transformer regressors.
- Evaluated models using MAE, RMSE, and R^2 metrics with comprehensive visualization and analysis tools.

Fractional Langevin Dynamics Simulator

 [GitHub](#)

- Developed a C++ simulator for active–passive particles in viscoelastic media using generalized Langevin dynamics.
- Implemented fractional Gaussian noise generation using the Hosking algorithm and computed particle trajectories and mean-squared displacement.
- Parallelized trajectory generation using MPI, enabling efficient large-scale ensemble simulations.

Molecule Generation using Deep Generative Models

 [GitHub](#)

- Implemented LSTM- and Transformer-based generative models for molecular SMILES generation using PyTorch.
- Built data preprocessing, training, sampling, and evaluation pipelines for molecular sequence generation.

SKILLS

Programming	Python (NumPy, SciPy, pandas, Matplotlib, PyTorch), C++, MATLAB, Mathematica
Machine Learning	Classical Machine Learning, PyTorch, Deep Learning, Graph Neural Networks (learning) Physics-Informed Neural Networks, Transformers, Diffusion Models, Variational Autoencoders, Time-Series analysis
Analytical & Computational Modeling	Stochastic Processes, Brownian Dynamics, Monte Carlo Simulation, Non-Markovian Dynamics, Statistical Inference
Scientific Computing	Numerical Optimization, Scientific Computing, High-Performance Computing (HPC)
Mathematics	Probability Theory, Statistics, Statistical Mechanics, Differential Equations

SCIENTIFIC TALKS AND POSTERS

- 2025 Delivered a scientific talk at *Chemsymphoria 2025*, IISER Pune, Pune, India.
- 2025 Poster presentation at *Inter IISER–NISER Chemistry Meet*, IISER Pune, India.
- 2024 Poster presentation at *Society of Physical Chemistry (SoPhyC) Conference*, Indian Institute of Technology Bombay, Mumbai, India.
- 2024 Poster presentation at *RSC Meet the Editors*, IISER Pune & CSIR–National Chemical Laboratory, Pune, India.
- 2024 Poster presentation at *International Conference on Fundamental and Advanced Research in Chemistry*, Indian Institute of Technology Mandi, India.
- 2023 Poster presentation at *Theoretical Chemistry Symposium*, Indian Institute of Technology Madras, Chennai, India.

PUBLICATIONS

Research Highlights: 9 peer-reviewed publications (8 first-author); 2 additional first-author manuscripts under revision.

1. **Jangid, P.;** Chaudhury, S. *Anomalous diffusion of coupled active-passive particles in viscoelastic media.* Under peer review (2026).
2. **Jangid, P.;** Chaudhury, S.; Kolomeisky, A. *Dynamic Fluctuations Can Counteract Catalytic Inhibition and Enhance Catalytic Performance.* Under peer review (2026).
3. **Jangid, P.;** Chaudhury, S. *DNA target search in chromatin compartments under stochastic resetting.* *J. Phys. Chem. B* **2026**, 130, 3766–3775.
4. **Jangid, P.;** Chaudhury, S. *Stochastic dynamics of nanoparticle catalysis: A discrete-state perspective.* *Mater. Horiz.* **2026**, 13, 2152–2172.
5. **Jangid, P.;** Metzler, R.; Chaudhury, S. *Internal protein motion in a rough model potential.* *J. Chem. Phys.* **2025**, 163, 244101.
6. **Jangid, P.;** Chaudhury, S.; Kolomeisky, A. *Effect of temperature in dynamic catalysis.* *J. Phys. Chem. C* **2025**, 129, 15606–15618.
7. **Jangid, P.;** Chaudhury, S.; Kolomeisky, A. *Theoretical understanding of dynamic catalysis.* *J. Phys. Chem. C* **2024**, 128, 9077–9089.
8. **Jangid, P.;** Chaudhury, S. *Transition Path Dynamics of Non-Markovian Systems across a Rough Potential Barrier.* *J. Phys. Chem. A* **2024**, 128, 10041–10052.
9. **Jangid, P.;** Punia, B.; Chaudhury, S. *Stochastic dynamics of hairballs in single-polymer growth.* *Phys. Chem. Chem. Phys.* **2024**, 26, 29749–29758.
10. Chaudhury, S.; **Jangid, P.;** Kolomeisky, A. B. *Dynamics of chemical reactions on single nanocatalysts with heterogeneous active sites.* *J. Chem. Phys.* **2023**, 158, 074101.
11. **Jangid, P.;** Chaudhury, S. *Exploring the role of heterogeneity in quorum sensing cells: A discrete-state stochastic approach.* *J. Stat. Mech.* **2023**, 2023, 093501.

AWARDS & FELLOWSHIPS

- 2026 ANRF-ITS Travel Grant, Anusandhan National Research Foundation (ANRF), India
- 2024 Best Poster Award, Royal Society of Chemistry – CrystEngComm Symposium, IISER Pune
- 2023 Best Teaching Assistant Award, Department of Chemistry, IISER Pune
- 2023 CSIR-UGC NET, Council of Scientific and Industrial Research (CSIR-UGC), India
- 2021 PhD Research Fellowship, IISER Pune
- 2021 Graduate Aptitude Test in Engineering (GATE), Indian Institute of Technology (IIT) Bombay, India
- 2020 Certificate of Merit, IISER Mohali
- 2016 Joint Entrance Examination (JEE Advanced), Indian Institute of Technology (IIT) Guwahati, India

TEACHING

Worked as a Teaching Assistant (TA) in *Quantum Chemistry, Organic Chemistry, Thermodynamics* at IISER Pune, India.